Polymeric Chemistry
Rheology

Universität Stuttgart

Author 1:	Laurens Viehoff
Author 2:	Felix Roth
E-Mail 1:	st183688@stud.uni-stuttgart.de
E-Mail 2:	st188157@stud.uni-stuttgart.de
Student-ID 1:	3676761
Student-ID 2:	3711192

Assistent:	Semi Kim

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1 Theory

In rheology, the flow behaviour of a substance under different forces is studied. This behaviour can be determined using rotational or oscillatory measurements.

1.1 Basic concepts

To measure rheological properties, the sample is placed between two plates, the upper one of which is capable of moving. When this plate is set in motion, the sample experiences a force and moves along with it, but exhibits a velocity gradient from the lower to the upper plate. A visual representation of this is shown in Figure 1.

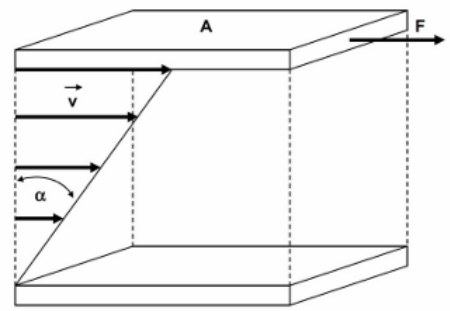

Figure 1: The diagram shows two plates with an area A, the upper one of which exerts a force F on the specimen, causing it to experience a velocity gradient $\vec{v}$ and to move through an angle α. [1]

Key parameters that can be detected or specified in this context include deformation γ, deformation rate $gam\dot{m}a$, shear stress τ and steady-state shear viscosity η. These parameters are defined by Equations (1–4).

$$\text{Deformation:} \qquad \gamma \equiv \frac{dx}{dy} = \tan\alpha \tag{1}$$

$$\text{Deformation rate:} \qquad \dot{\gamma} \equiv \frac{d\gamma}{dt} \tag{2}$$

$$\text{Shear stress:} \qquad \tau = \frac{F}{A} = \eta \cdot \dot{\gamma} \tag{3}$$

$$\text{Steady-state shear viscosity:} \qquad \eta = \frac{\tau}{\dot{\gamma}} \tag{4}$$

x and y represent the coordinate axes, with x pointing in the direction of the force F.
If a strain rate is specified and the shear stress is thereby detected, the viscosity can also be determined, allowing these parameters to be investigated as a function of the strain rate. Three types of curve behaviour can be detected for the shear stress or viscosity. The parameters may follow a linear trend or deviate from it. If the parameters deviate, they may become either larger or smaller at higher deformation rates. If a sample exhibits a linear relationship, it is referred to as a Newtonian fluid. If the parameters increase at higher deformation rates, it is referred to as a non-Newtonian shear-thickening fluid; conversely, if they decrease, it is referred to as a non-Newtonian shear-thinning fluid. The curves can be seen in Figure.

2 Results

2.1 Rotational measurements

After analyzing the different graphs it can be stated that honey has the typicall linear progression wich speaks for a newtonian fluid. By analyzing Graph 2 in the area from 10^{-1}-10^{0} of the deformation velocity a non linear part can be observed. However, this does not mean that honey is a non-Newtonian fluid, but rather that it takes a certain amount of time before measurement accuracy can be guaranteed. This comes from the first "loosening" of the molecules, because these can be intertwinded or stuck in each other. This can also be applied on the other measurements.

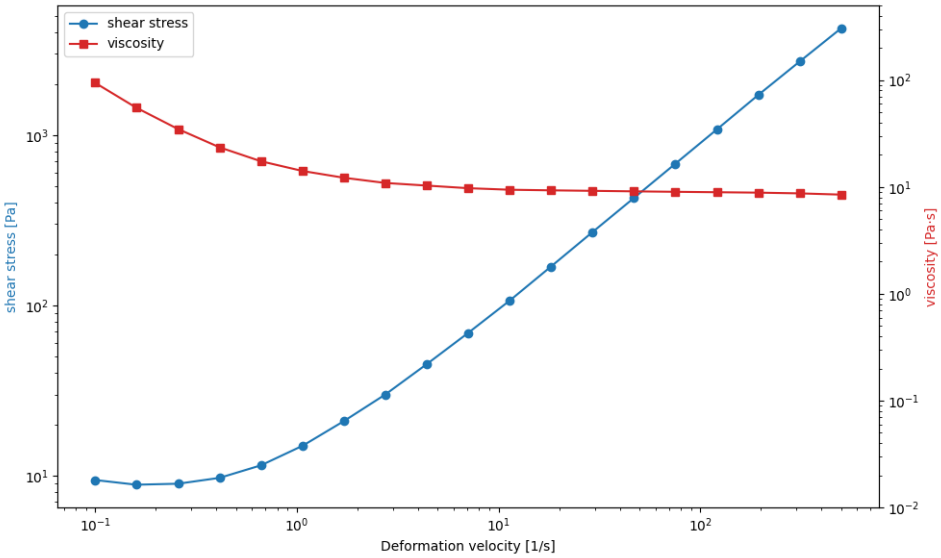

Figure 2: The graph shows the measurements of honey. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

By applying the same procedure from the honey a clear trend for the 14%-PVP solution can be observed. Graph 3 shows a tiny decline in the end wich implies a share thinning non newtonian fluid. Meaning it has a viscosity decrease with increasing deformation velocity. This is also confirmed after research [2].

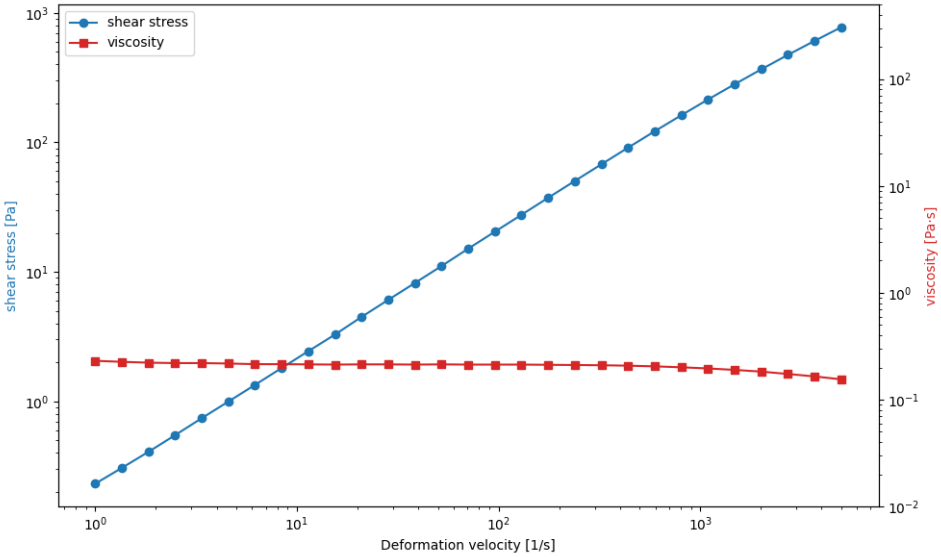

Figure 3: The graph shows the measurements of 14%-PVP solution. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

The final measurement of this exercise was the measurement of a 50% starch solution. Graph 4 shows a clear exponential funktion wich points to a non newtonian shear thickening fluid. Meaning the viscosity is increasing when increasing the deformation velocity. This is not that surprising because starch is also the main ingredient of oobleck and the properties are pretty known.

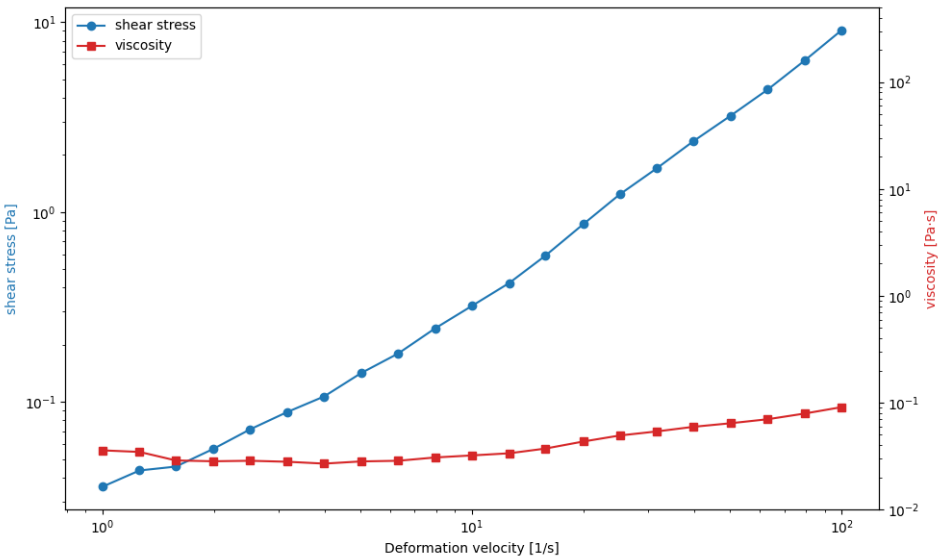

Figure 4: The graph shows the measurements of a starch solution. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

2.2 Time dependant measurement of the viscosity

The next task was to determine the viscosity of ketchup in the dependency over time. This measurement was repeated for three times under the same conditions. After examination of

Graph 6, a difference between the viscosity before and after applying the deformation velocity is visible. This behaviour is typicall for

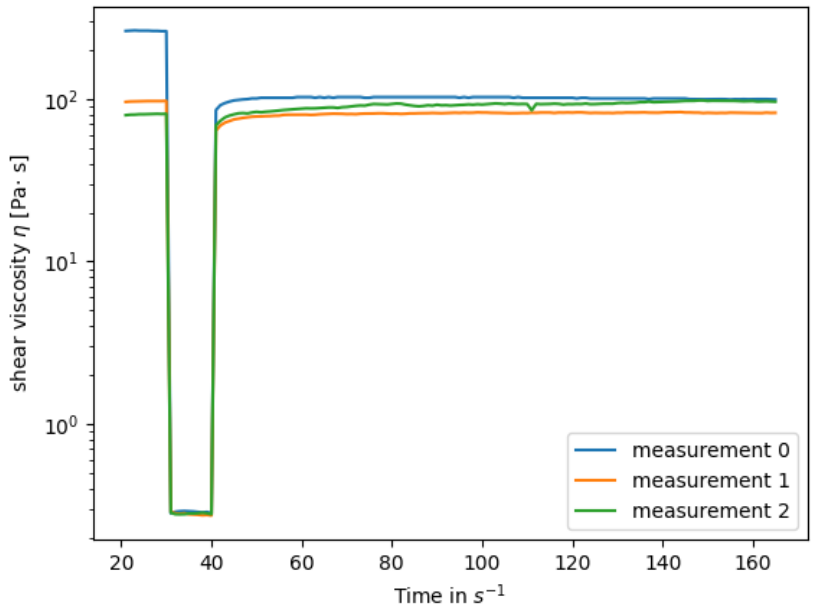

Figure 5: The graph shows the three time dependant viscosity measurements of ketchup. The shear viscosity is plotted against the duration.

2.3 Oszillational measurements to determine the viscoelastic properties

2.3.1 Amplitude sweep

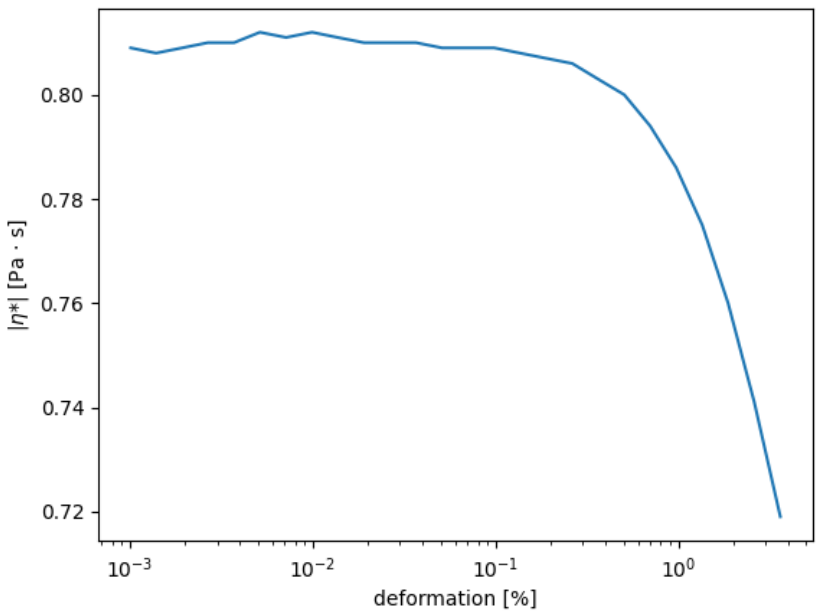

Figure 6: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant frequency.

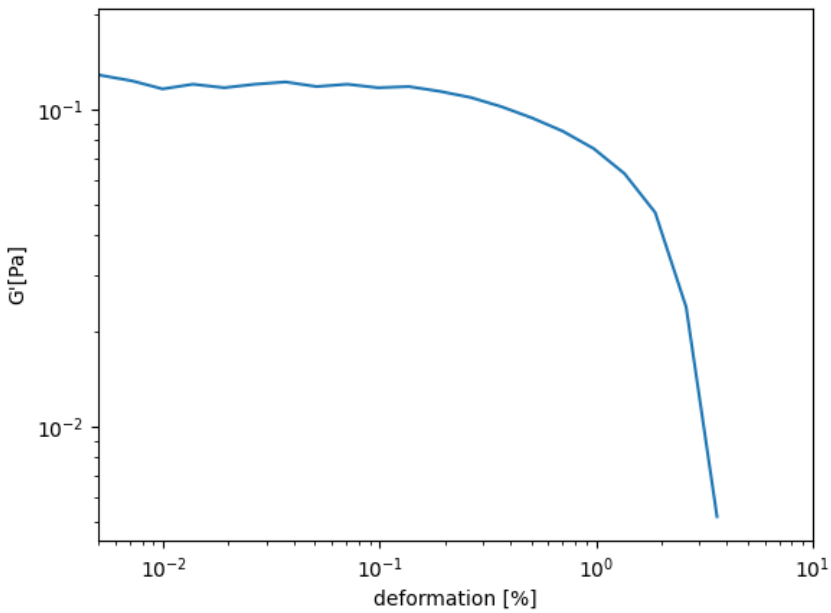

Figure 7: This graph shows the storage module G' plotted against the deformation of 20%-PVP solution at constant frequency.

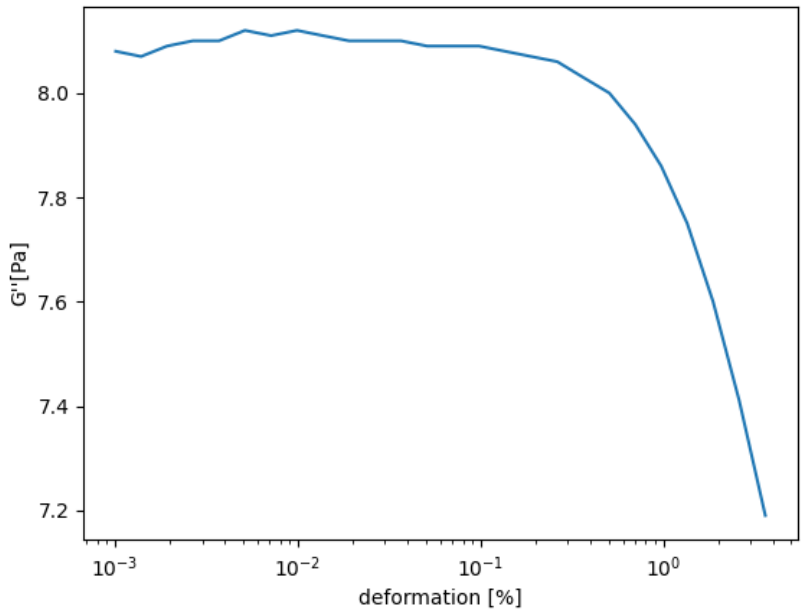

Figure 8: This graph shows the loss module G" plotted against the deformation of 20%-PVP solution at constant frequency.

2.3.2 Frequency sweep

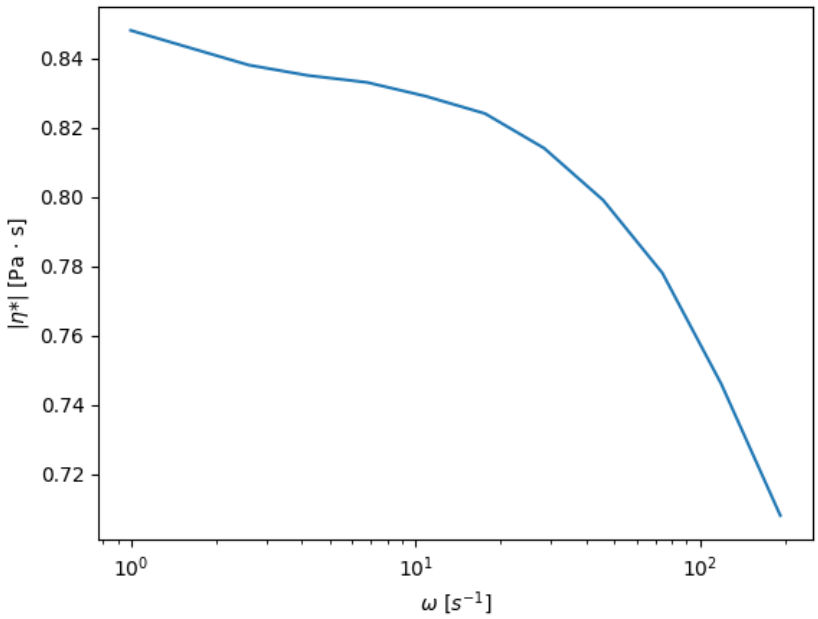

Figure 9: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant amplitude.

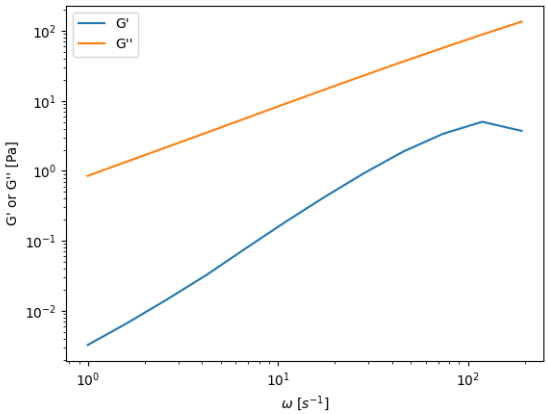

Figure 10: This graph shows the storage module G' loss module G" plotted against the deformation of 20%-PVP solution at constant amplitude.

References

[1] Institut für Polymerchemie - Polymerpraktikum, Vorlesungsskript, Universität Stuttgart, **März 2025**.

[2] J. E. Martín-Alfonso, E. Číková, M. Omastová, *Composites Part B: Engineering* **2019**, *169*, 79–87.